

Empirical Study of Oversmoothing in Citation Graph Neural Networks

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Abstract

Graph neural networks on dense citation networks experience exponential feature similarity decay when layer depth increases beyond threshold $K=4$.

1. Introduction

Analyzing scientific literature using graph representations has revolutionized bibliometrics. However, multi-hop aggregation introduces severe representation collapse.

2. Limitations

Our experiments are strictly limited to homogeneous citation graphs and do not generalize to heterogeneous author-institution networks.

References

[1] Thomas N. Kipf and Max Welling. Semi-supervised classification with graph convolutional networks. ICLR 2017.